# MANUEL BUENROSTRO MACEDO

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Engineering Portfolio: https://buenrosm23.github.io/

## EDUCATION

Embry-Riddle Aeronautical University — Daytona Beach, FL
Bachelor of Science in Aerospace Engineering, Rocket Propulsion
Expected Graduation: May 2028 | GPA: 3.0 / 4.0
Relevant Coursework: Engineering Fundamentals, Engineering Graphics (CATIA V5), MATLAB Programming, Calculus II & III, Physics, Chemistry and Lab

## ENGINEERING EXPERIENCE

**Engineering Physics Propulsion Lab (EPPL)** — Embry-Riddle Aeronautical University
Undergraduate Research Assistant | 2024 – Present
- Collaborated in mechanical integration and iteration testing of lab hardware.
- Presented technical solutions to sponsors, visitors, and faculty at lab showcase events.
- Applied engineering design principles to create, revise, and improve mechanical parts in Autodesk Inventor.
- Gained firsthand experience in a professional research lab setting.

## TECHNICAL PROJECTS

**OpenArms – Open-source Robotic Arm — EPPL, ERAU | January 2025 - Present**
- Team member in the design of an open-source robotic arm attachment for OpenMutt.
- Responsible for CAD modeling and developing modular end-effectors.

**OpenMutt – Open-source Quadruped — ERAU | September 2024 – Present**

- Designed and fabricated a pan-tilt sensor mount to extend quadruped functionality using 3D printing and Autodesk Inventor.
- Designed and fabricated a quick disconnect mounting system for interchangeable end-effectors extending quadruped functionality using 3D printing and Autodesk Inventor

**VEX Robotics Competition – Team 7121E | March 2020 – May 2024**
- Led 7-member team to 21 regional, 3 state, and 1 world championship appearances; won 7 design-based awards.
- Team leader, head designer; built award-winning robots with pneumatics and CAD; maintained detailed engineering documentation.

## TECHNICAL & PROFESSIONAL SKILLS

**Software & Tools:** Autodesk Inventor, CATIA V5, MATLAB (basic), Python (beginner), 3D printing, laser cutting, CNC (beginner), Windows Ecosystem, Visual Studio Code (Beginner), Prusa Slicer, Bambu Studio, Cura.

**Soft Skills:**
-Leadership, teamwork, problem-solving, public speaking, technical presentations, technical report writing
-Experience with shop tools, and safety procedures.
-Languages: Fluent in English and Spanish